Chapter 4

Making Custom Copilots

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» **Understanding Copilot agents**

» **Using Copilot Studio**

» **Prompting for agents**

» **Giving agents knowledge**

» **Adding new topics to agents**

» **Editing agents**

» **Publishing Copilot agents**

Throughout this book, you've seen and explored many ways that generative AI (GenAI) has been integrated into different applications and types of devices. Thus far, I've only touched on a small subset of everything GenAI chatbots are capable of. My favorite feature of the latest GenAI chatbots is the ability to create customized chatbots to handle specific tasks by referencing documents or knowledge sources that you provide.

Microsoft Copilot Studio is a tool that enables you to create your own *copilots* (also known as Copilot agents). As you find out in this chapter, Copilot Studio is a powerful tool that's also easy to use.

Building Your Own Copilot Agent with Copilot Studio

Microsoft Copilot Studio lets you create your own Copilot agents or Copilot plugins and then publish them to be used in Microsoft 365 Copilot.

To try out Copilot Studio, go to `www.microsoft.com/en-us/microsoft-copilot/microsoft-copilot-studio` in your browser. You'll see a screen similar to the one shown in Figure 4-1.

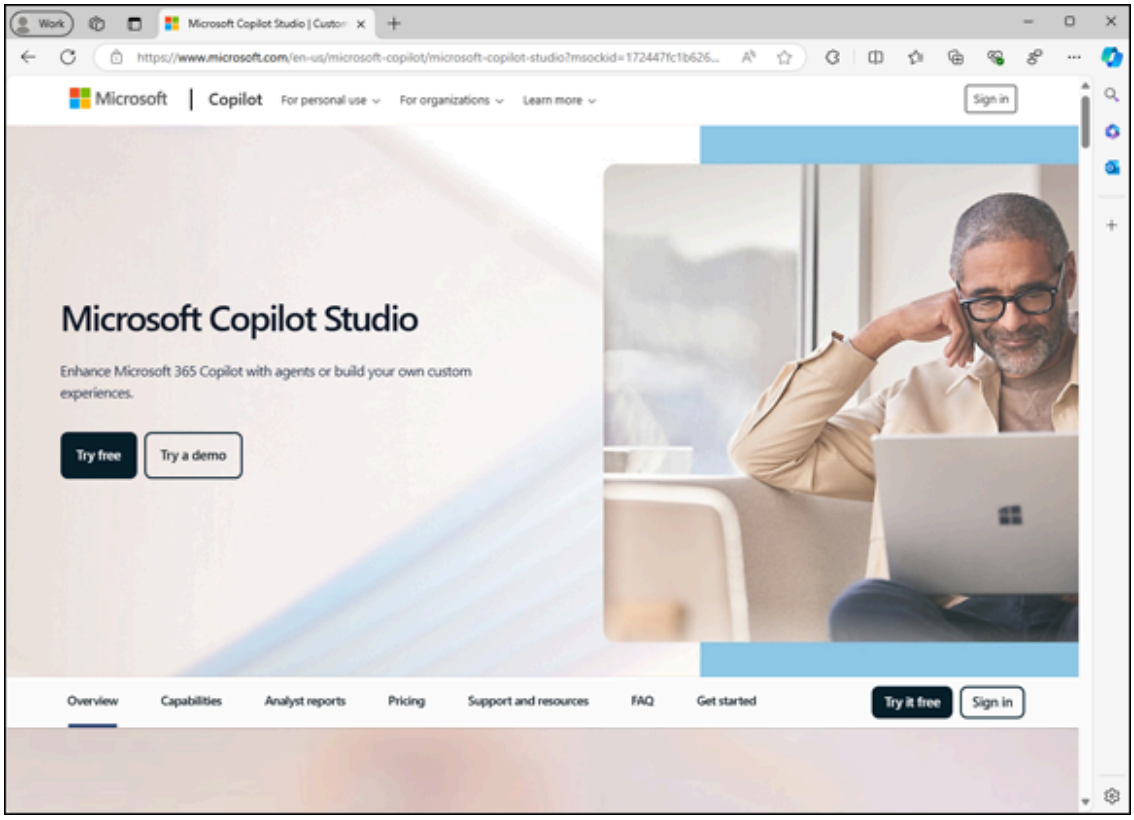


FIGURE 4-1: The Copilot Studio homepage.

Click the Try a Demo button. This will take you to the Copilot Studio demo site at `https://copilotstudio.microsoft.com/tryit`, where you can enter any website address and create a Copilot agent that can answer questions specifically about that website.

When you first open the Copilot Studio demonstration, a popup window will appear, as shown in Figure 4-2, and ask you for the URL of your website.

If you don't have your own website, you can enter any other website you like here. I entered my website address (`www.chrisminnick.com`).

After you enter a website address, your agent will be created, and it will greet you as the *virtual assistant* for the website you entered, as shown in Figure 4-3.



TECHNICAL STUFF Behind the scenes, Copilot Studio prefaces the conversations in this window with an instruction to use search results from Bing for the website address you entered to answer questions.

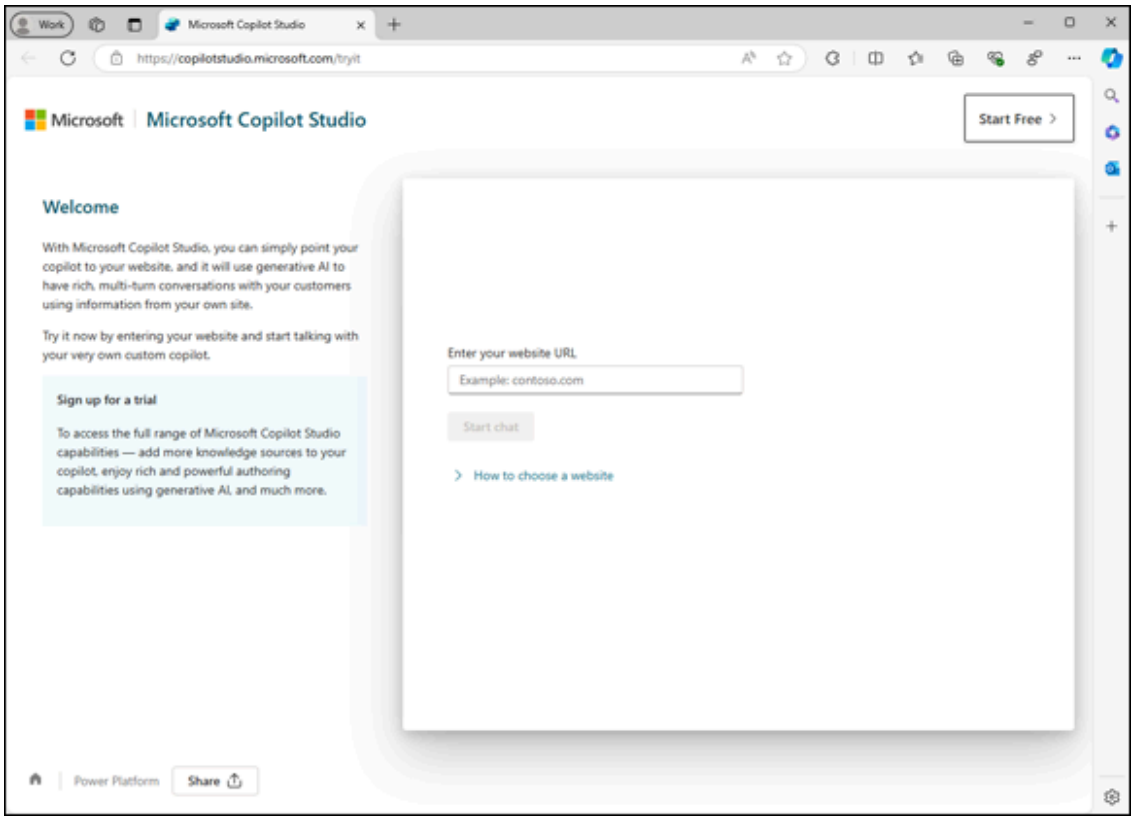


FIGURE 4-2: Getting started with the Copilot Studio demo.

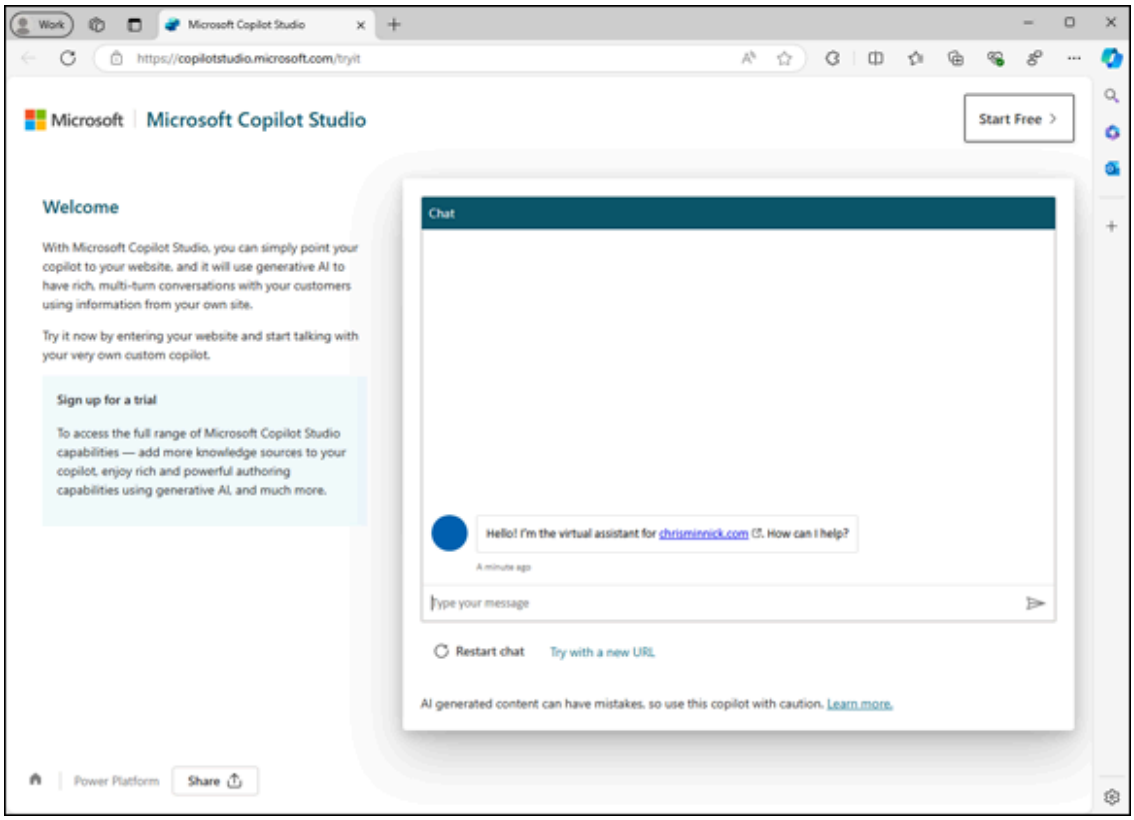


FIGURE 4-3: Meet your virtual assistant.

In my tests of this demo, my virtual assistant for `chrisminnick.com` was able to reliably return correct information from my website along with references for the exact pages where it got the information. Figure 4-4 shows a typical response.

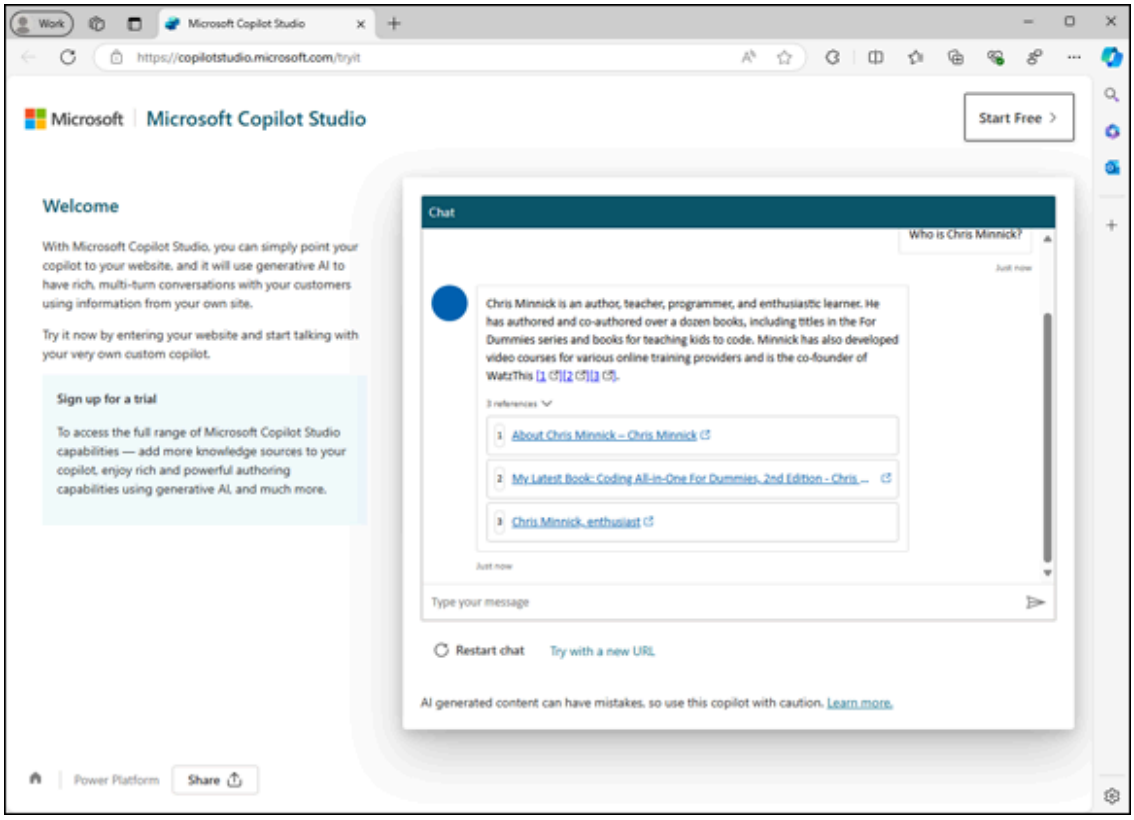


FIGURE 4-4: A response from the `chrisminnick.com` virtual assistant.

You've likely seen this type of AI customer service agent before. Check out Chapter 3 in this book to find out more about how and why businesses create Conversational AI (CAI) chatbots. Now that you know how easy it is to create one of these agents, you can try out some more complex projects. Click the Start Free button in the upper-right corner of the demo screen.

If you're logged into a Microsoft account with access to Copilot Studio, you'll be asked if you want to use Copilot Studio with that account. Otherwise, you'll be asked to log in or create a new account.

After you've logged in to your Microsoft account, you'll be taken to Copilot Studio, where you'll see a Welcome slideshow, as shown in Figure 4-5.

When you close the welcome message, you'll see the Copilot Studio home-page, which will look similar to the screenshot in Figure 4-6.

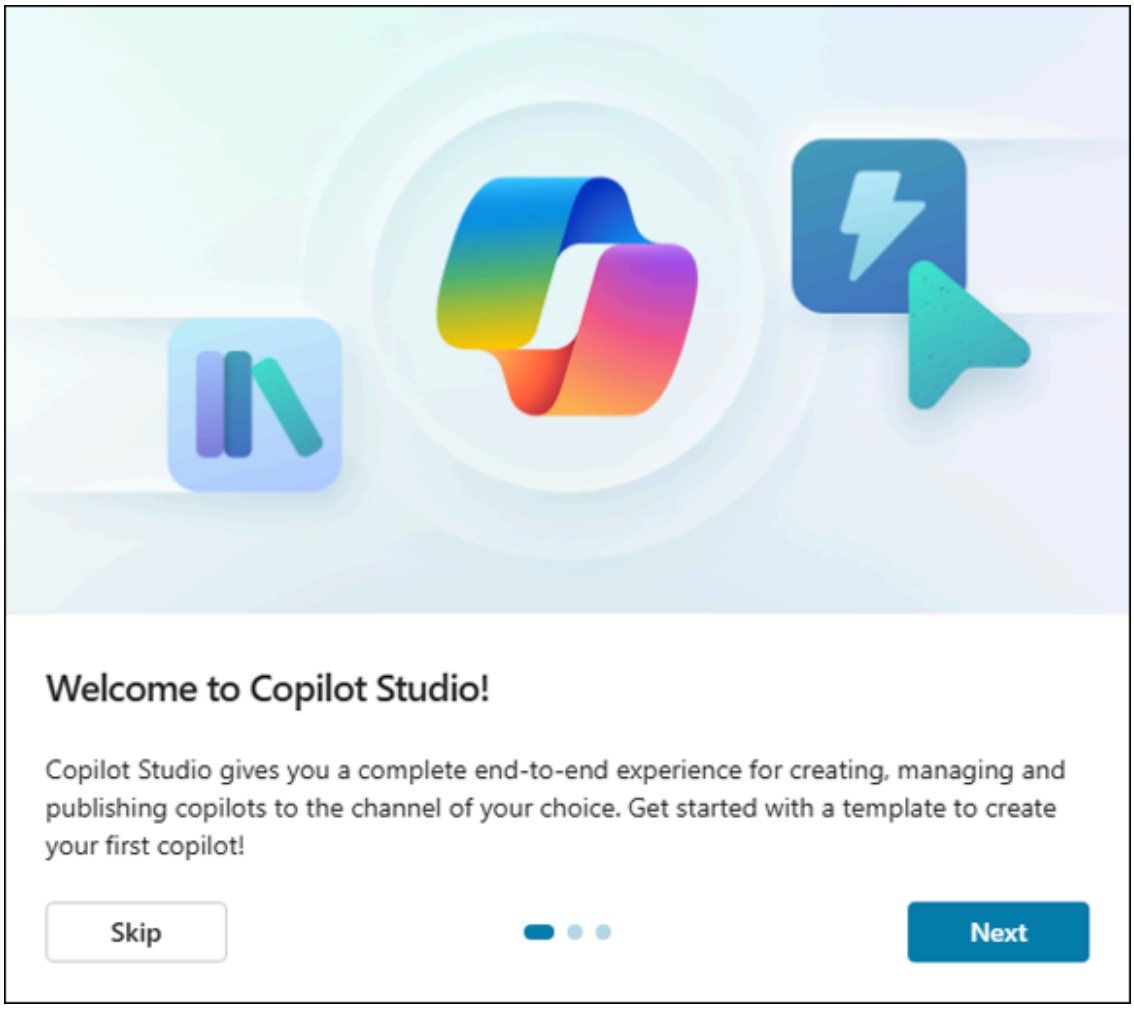


FIGURE 4-5: The Copilot Studio welcome slideshow.

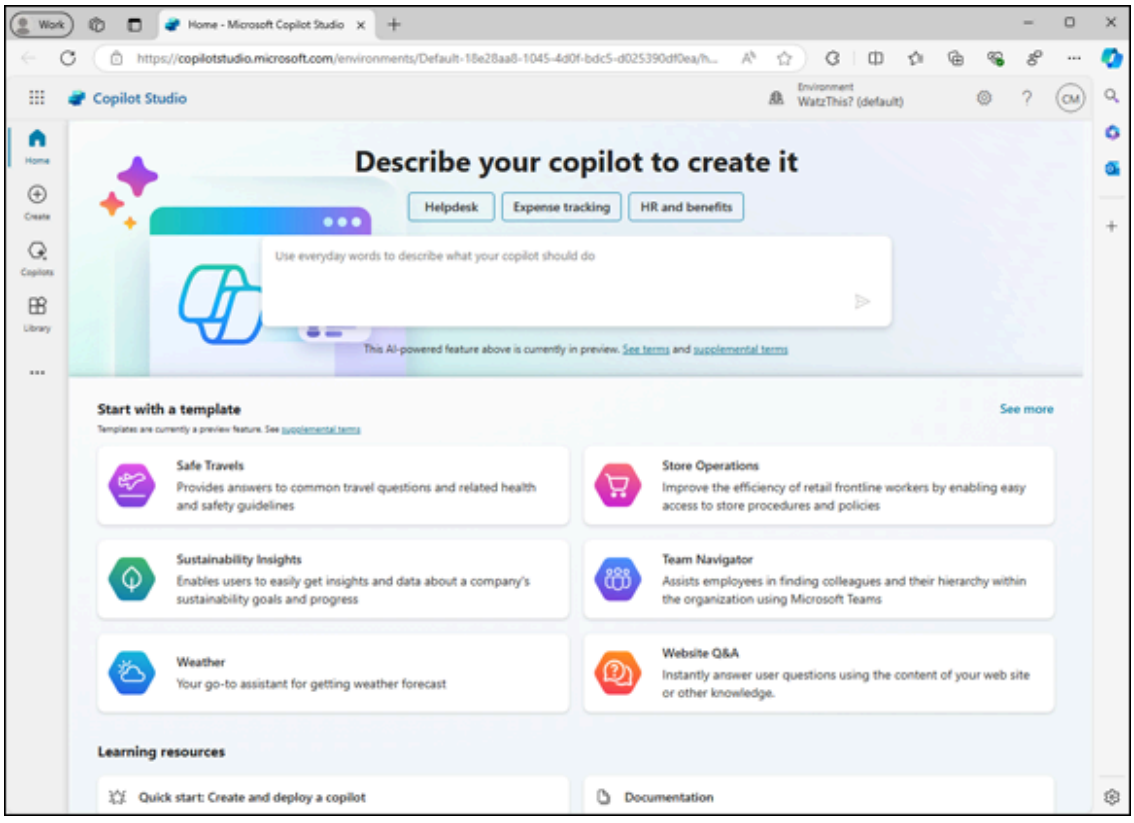


FIGURE 4-6: The Copilot Studio homepage.

You can create Copilot agents in one of two ways:

- Entering a description of the agent into the text input box in the top third of Copilot Studio
- Starting with a template

In the following sections, you will learn about these tools and the capabilities of each.

Creating agents by prompting

The prompting area in Copilot Studio doesn't offer a lot of help with how to use it other than the following instruction:

Use everyday words to describe what your copilot should do.

After thinking about it for a moment, I decided to start with something seemingly simple (for an AI agent, anyway). I gave Copilot Studio the following prompt:

> *A helpful assistant that references* `Wikipedia.org` *when answering questions.*

After submitting the prompt, a conversation window opens, as shown in Figure 4-7, and Copilot asks follow-up questions about the agent you want to create.

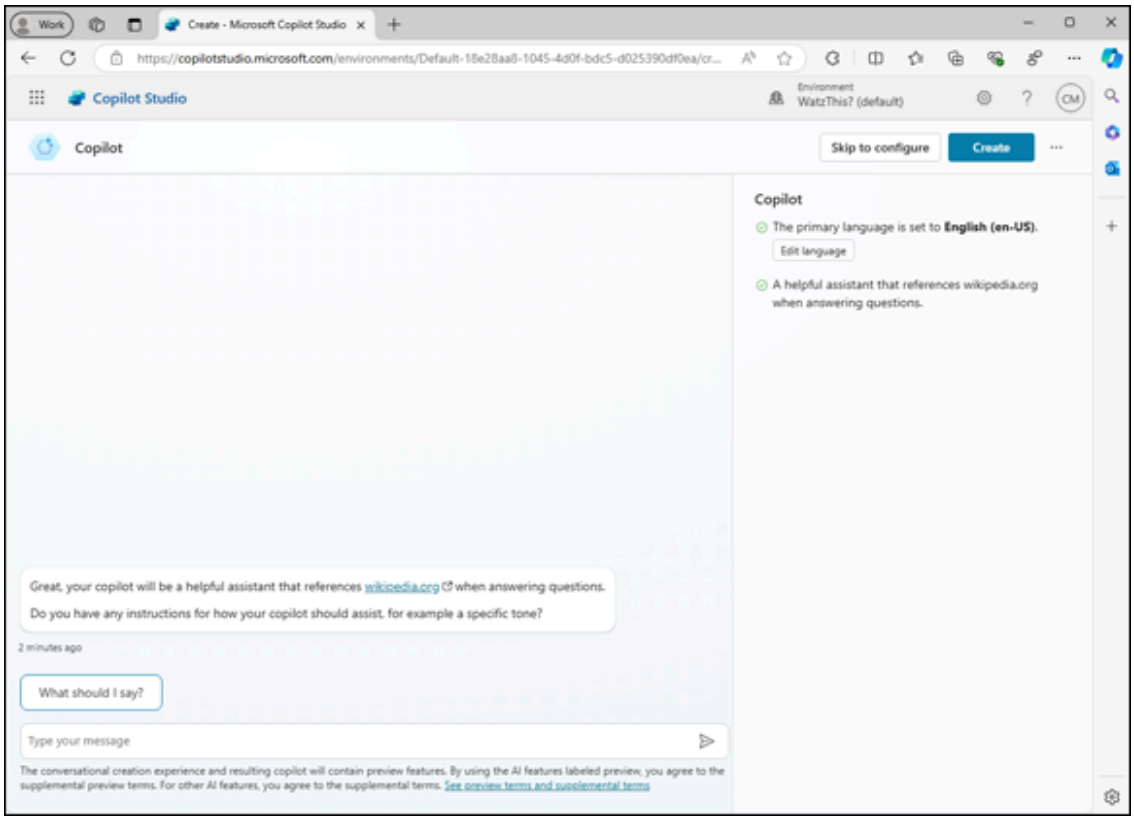


FIGURE 4-7: Chatting with the agent creator.

The follow-up questions it asks are all useful things to think about when you create an agent using the more manual process later on, including

- Do you have any instructions for how your copilot should assist, for example a specific tone?
- Where should the copilot find important information? Provide any publicly accessible websites that your copilot will need.
- Are there any topics or tasks this copilot shouldn't help with or talk about?

After you've answered the follow-up questions, you'll see the instructions that Copilot generated from your guidance and a button labeled I'm Done, as shown in Figure 4-8.

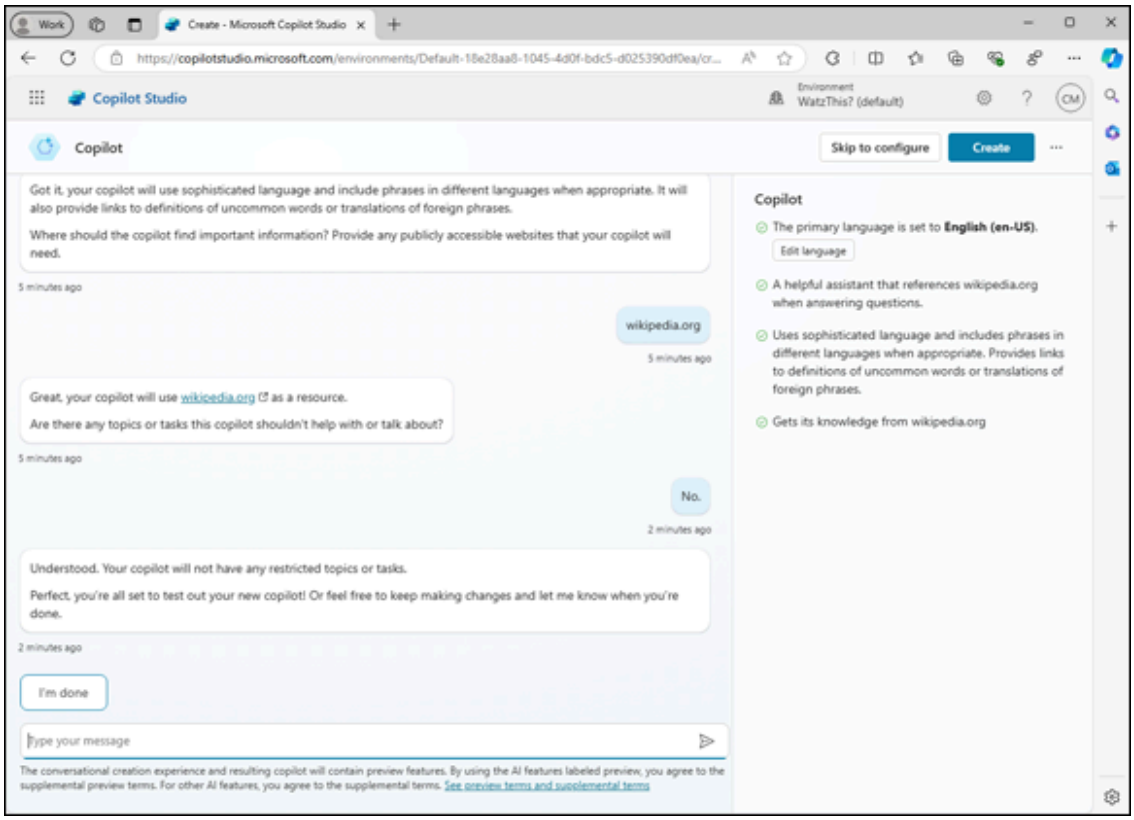


FIGURE 4-8: Completing the AI-assisted agent creation process.

If you're ready, click the I'm Done button to enter a conversation with the agent you've created. In my first test of the AI agent-creation process, I also had to click the Create button in the upper-right corner of the screen to complete the AI-assisted portion of the agent-creation process.

The next screen you'll see is shown in Figure 4-9.

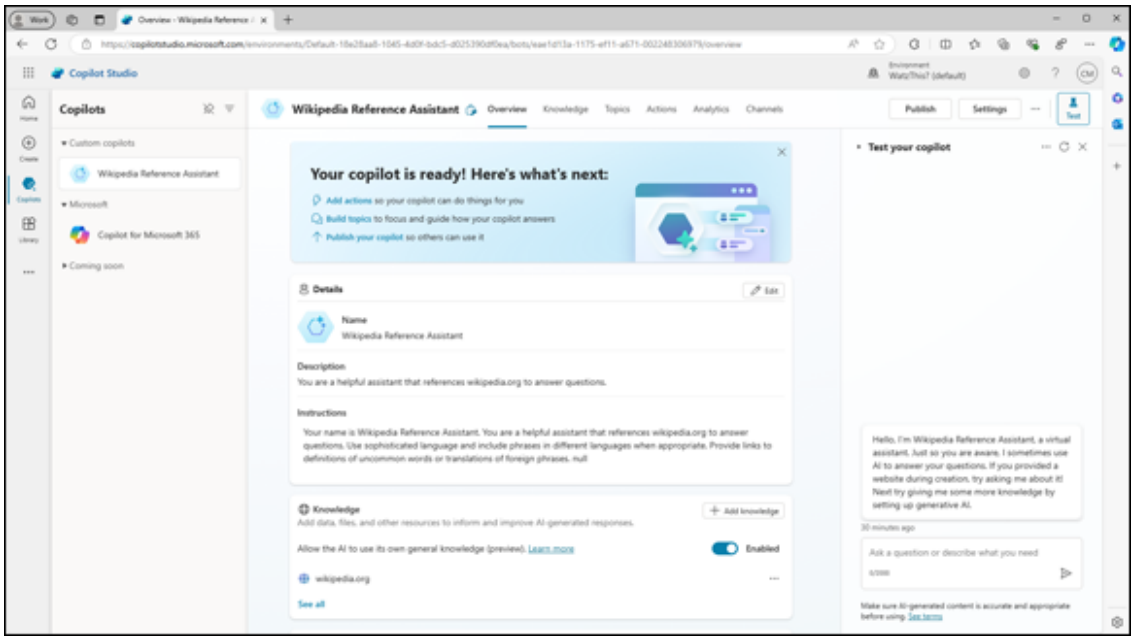


FIGURE 4-9: The agent editor.

Creating agents with templates

In addition to using the prompting method to create Copilot agents, you can also start with one of the pre-built templates. Ten templates are currently available, and another six are listed as Coming Soon on Copilot Studio's Create page.

To see the available templates, click the Create icon on the left side of Copilot Studio and scroll down to the section labeled Start with a Template. You'll see the options shown in Figure 4-10 (and hopefully additional ones, too!).

You can select any of the templates by clicking on it. A screen appears with a link to the template instructions and the template settings. Figure 4-11 shows the Website Q&A Copilot template instructions.

You can customize the name, description, instructions, or knowledge for the agent on this page; then click the Create button to create the agent. You'll then be taken to the Copilot Agent Editor interface.

From this point on, working with an agent started from a template is the same as working with any other agent. You can modify the agent's various settings, test it out, view the agent's analytics, and publish it when you're ready to share it with your selected channels. The next section explains how to do all this and more with your agent.

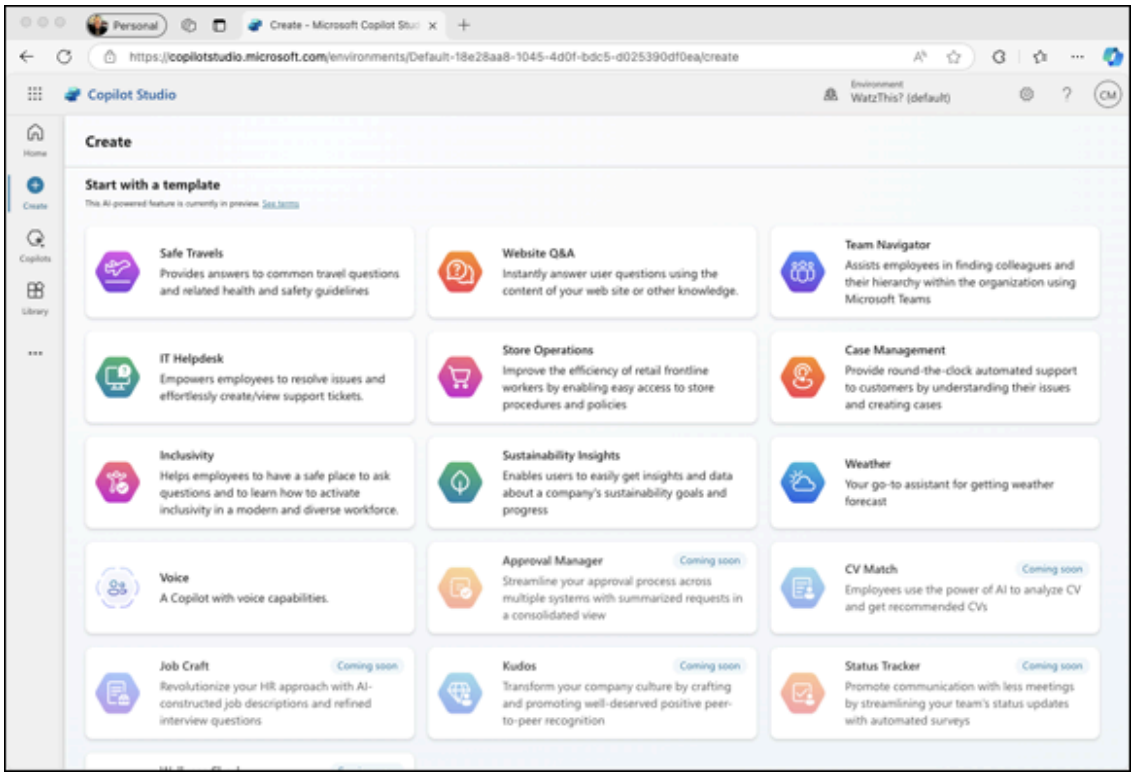


FIGURE 4-10: The available agent templates.

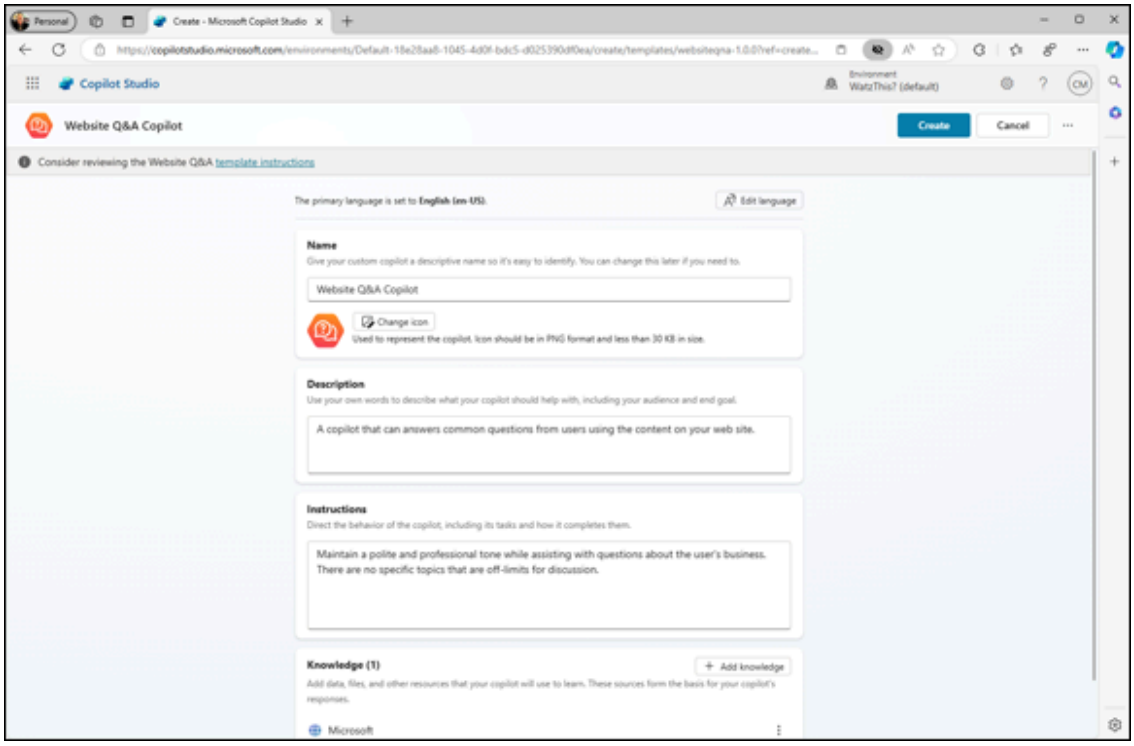


FIGURE 4-11: The Website Q&A template settings.

Testing and Editing Your Agent

The first thing you should do is to try out your new agent. Type a request into the Test Your Copilot sidebar and see what you think of the response.

By default, your agent will likely be pretty dry and boring — just the facts. Figure 4-12 shows my first conversation with my Wikipedia agent.

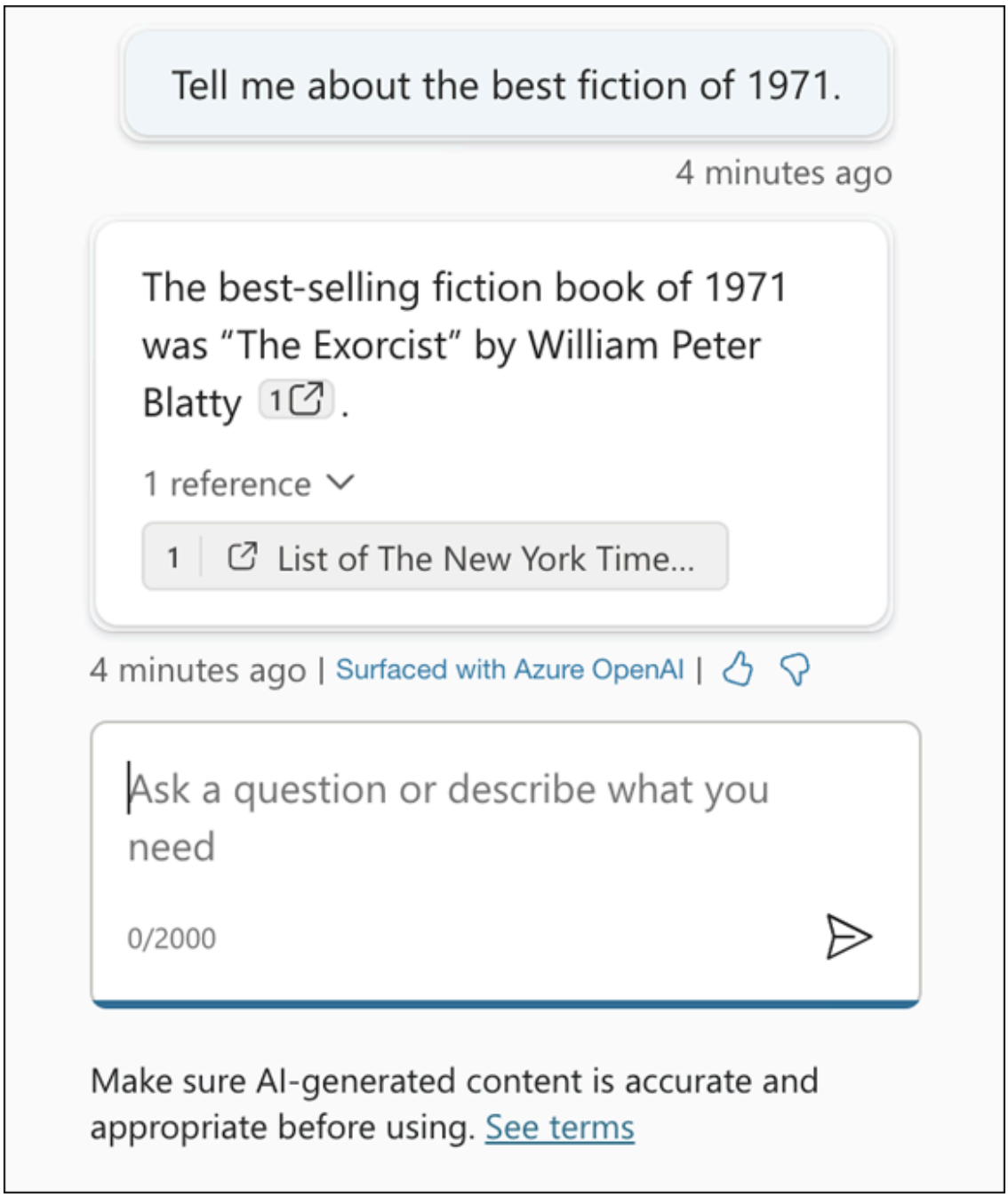


FIGURE 4-12: Copilot is sticking to the facts.

If this factual and dry approach is what you want, that's excellent. But, for my agent, I want to make it a bit more fun. To accomplish this, I need to adjust some settings.

Configuring generative AI

By default, your agent is configured to be strict about what particular phrases trigger its actions. It will use generative AI as little as possible

and will instead stick to using quotes and precise interpretations from the knowledge sources you provide. If you enable generative AI, your agent will be able to respond to requests without requiring an exact phrase and will be more creative.

To enable generative AI, click the Settings button in the upper-right corner and then click the Generative AI option on the left. You'll see a screen where you can read about the difference between the Classic mode and the Generative mode. You can select how strict the content moderation should be for your agent. In the case of this agent, I want it to be as creative as possible, so I enabled Generative mode and set content moderation to low strictness.

After you've made your selections and clicked the Save button, you can close the Settings. You'll be returned to the Copilot Agent Editor.

Using the Copilot Agent Editor

The Copilot Agent Editor is where you can adjust everything that makes your agent act the way it does. At the top of the interface is a row of tabs that allow you to customize different aspects of your AI agent. The first tab, Overview, displays a condensed version of the information from the other five tabs. These additional tabs are

- Knowledge
- Topics
- Actions
- Analytics
- Channels

The Knowledge tab

On this screen, shown in Figure 4-13, you can view a list of the knowledge sources your agent has access to. You can also add knowledge sources.

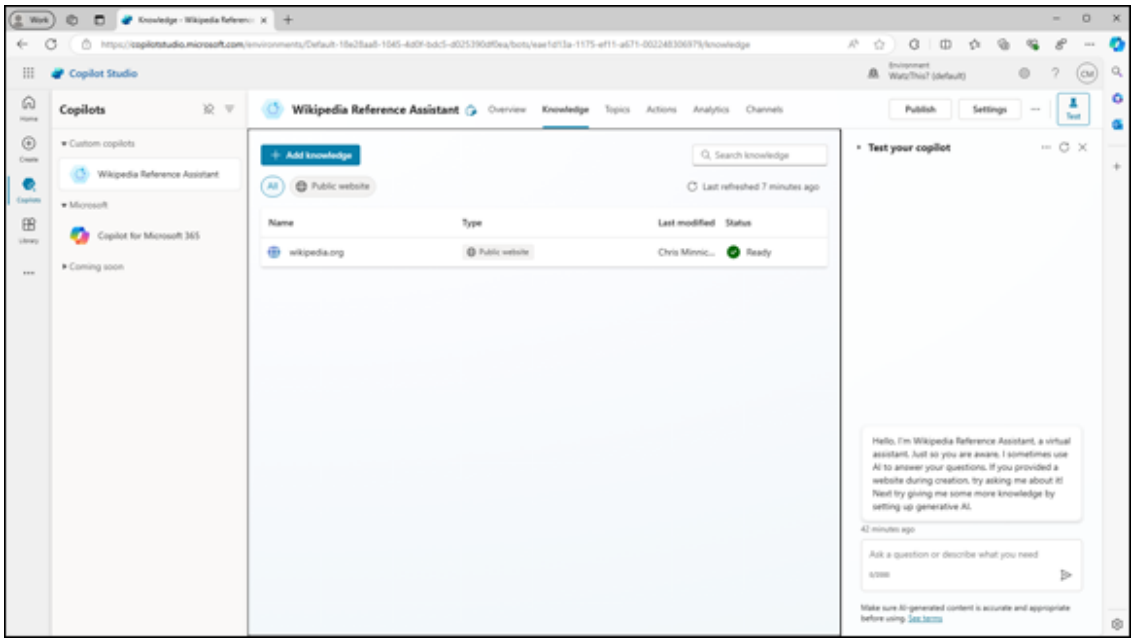


FIGURE 4-13: Viewing and editing your agent's knowledge.

Thus far, you've only added public websites as knowledge sources. By clicking the Add Knowledge button, you can give your agent access to data in specific files, data in SharePoint and OneDrive, data in enterprise databases, and much more. There's also a Custom Connector option where you write your own connector for a data source that isn't already covered by one of the other connectors.

Some of the connectors that are available by clicking the Add Knowledge button are shown in Figure 4-14.

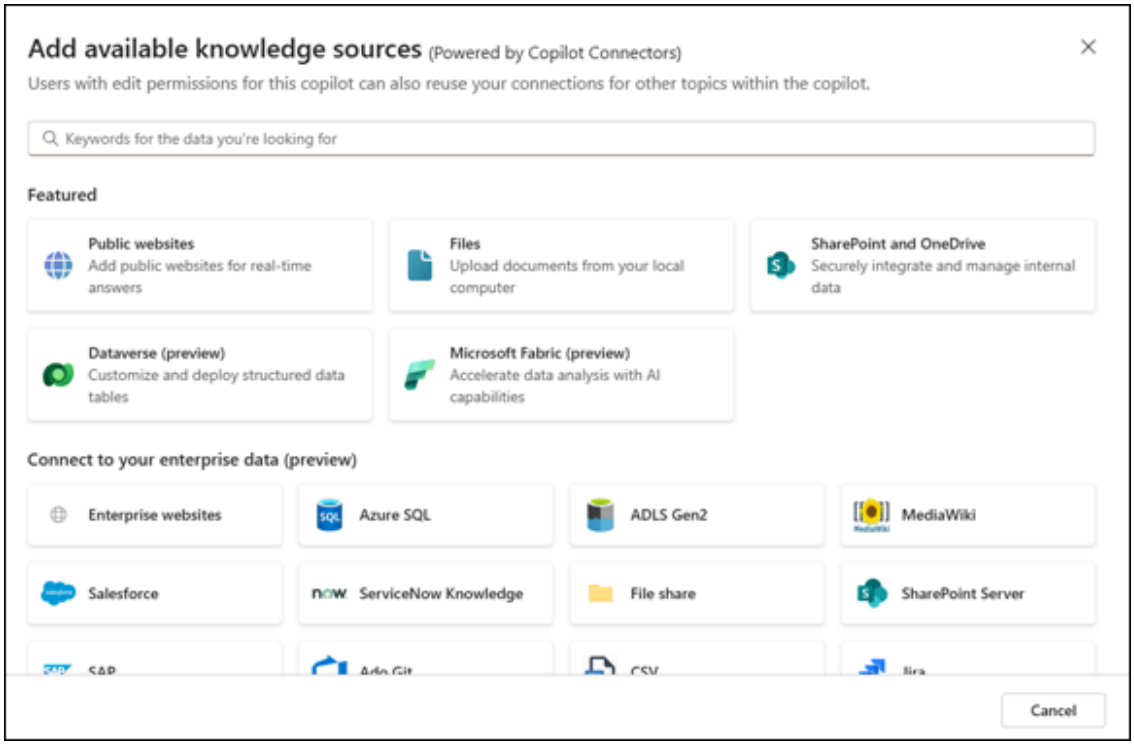


FIGURE 4-14: Viewing available connectors.

The Topics tab

The Topics tab, shown in Figure 4-15, is where you can configure the custom and system topics for your agent.

Topics are ways that the Copilot agent will respond when it detects a specific trigger. Topics are split into two categories: custom topics and system topics. Custom topics are topics that you've created (or that your AI agent creator assistant created) and that are specific to this agent. System topics are things that are built into Copilot and that you can customize for your agent.

If you switch to viewing the system topics, you can hover over each one to see what it does. For example, the On Error system topic triggers when the agent detects an error. If you want your agent to reply in some particular way when an error occurs, you can customize its response by clicking on the On Error system topic.

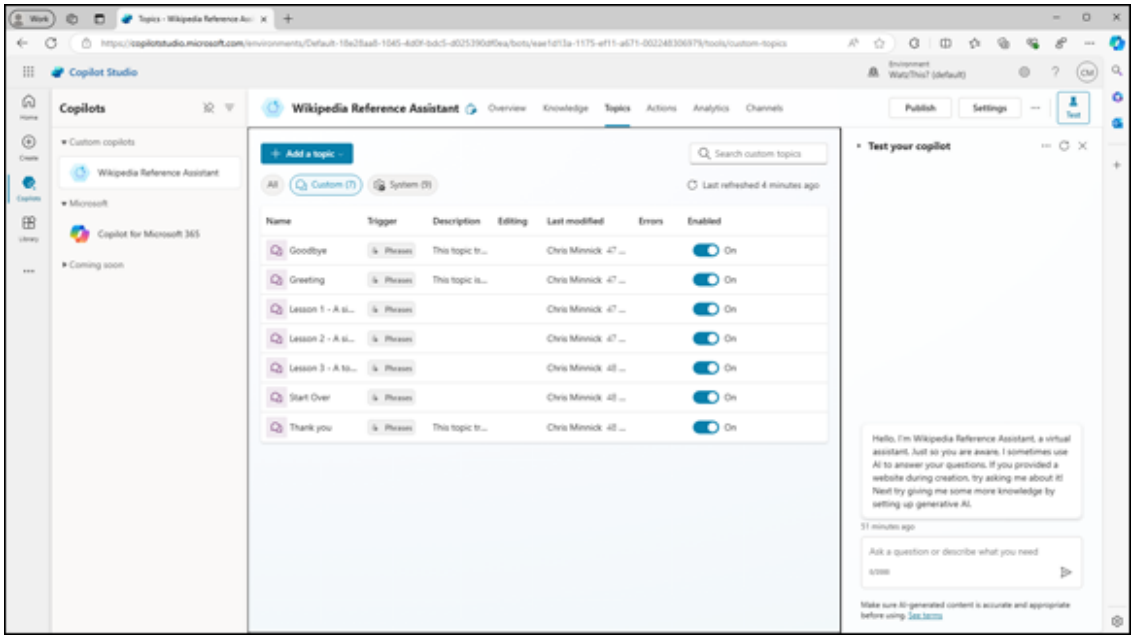


FIGURE 4-15: The Topics tab.

When you edit a topic, you'll see a flow chart view of what the topic does, as shown in Figure 4-16.

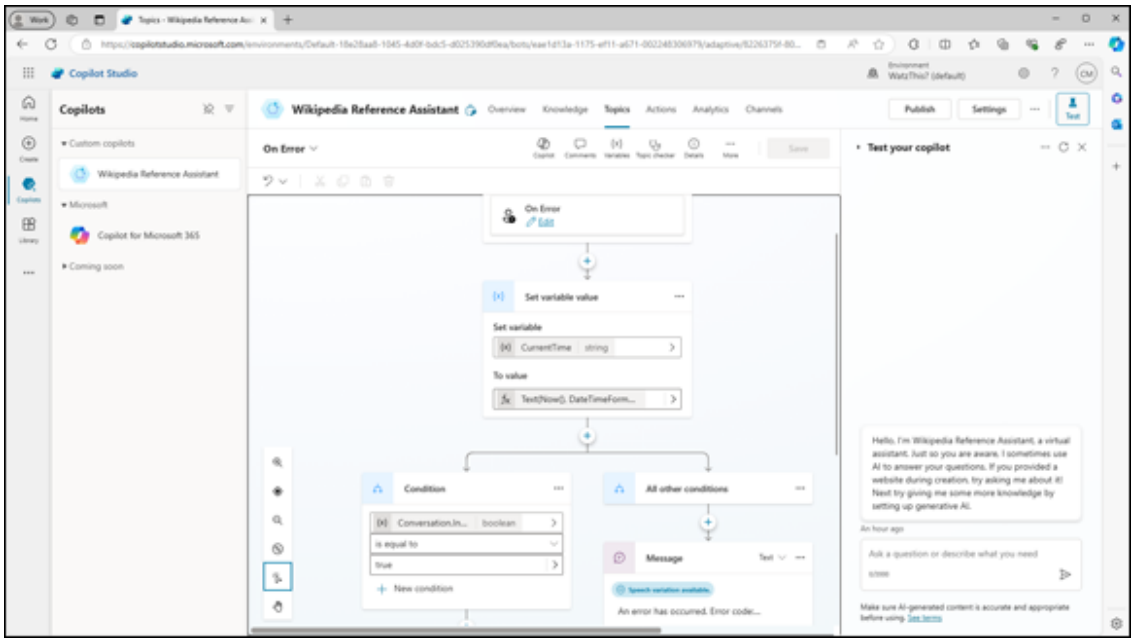


FIGURE 4-16: Editing a topic.

Every topic starts with a trigger. The *trigger* is the phrase or event that causes the topic to be activated. After the trigger, topics may involve multiple conditions, messages, actions, and more.

CREATE A TOPIC

The best way to understand what's possible with topics is to create a new one. As with creating entire agents, you can create new topics through prompting. Follow these steps:

1. **Click the Add a Topic button in the Topics tab.**
   A submenu will appear, as shown in Figure 4-17, where you can choose whether to create the topic from scratch or by entering a description.

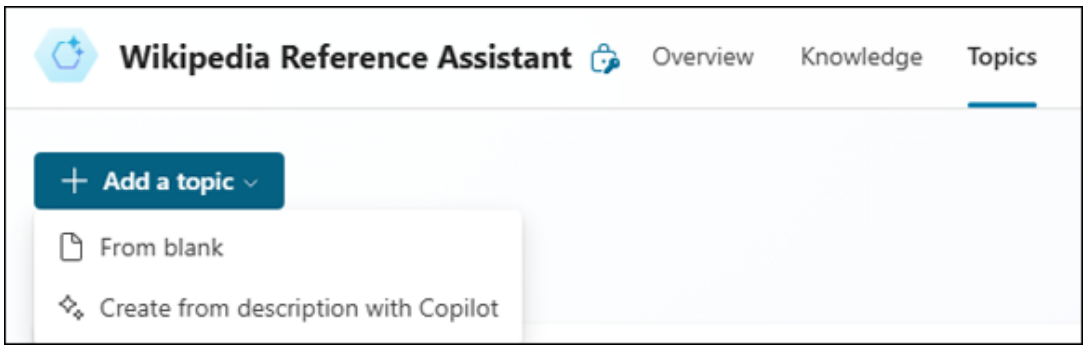


FIGURE 4-17: Choosing how to add the topic.

2. **Select Create from Description with Copilot.**
   The window shown in Figure 4-18 will open.

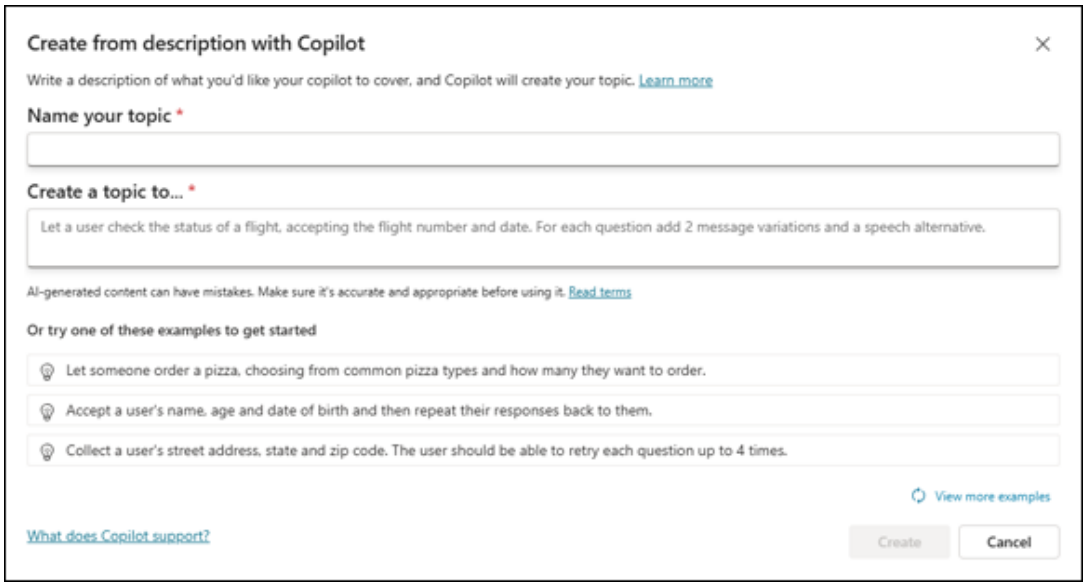


FIGURE 4-18: Describing your topic.

3. **View the examples on this page and select one by clicking on it or enter your own custom description.**
4. **Give your new action a name and click the Create button.**

Your new action will be created, and you'll see the topic editor, where you can see what Copilot added and make changes or additions to it. Figure 4-19 shows the user data collection action that I created.

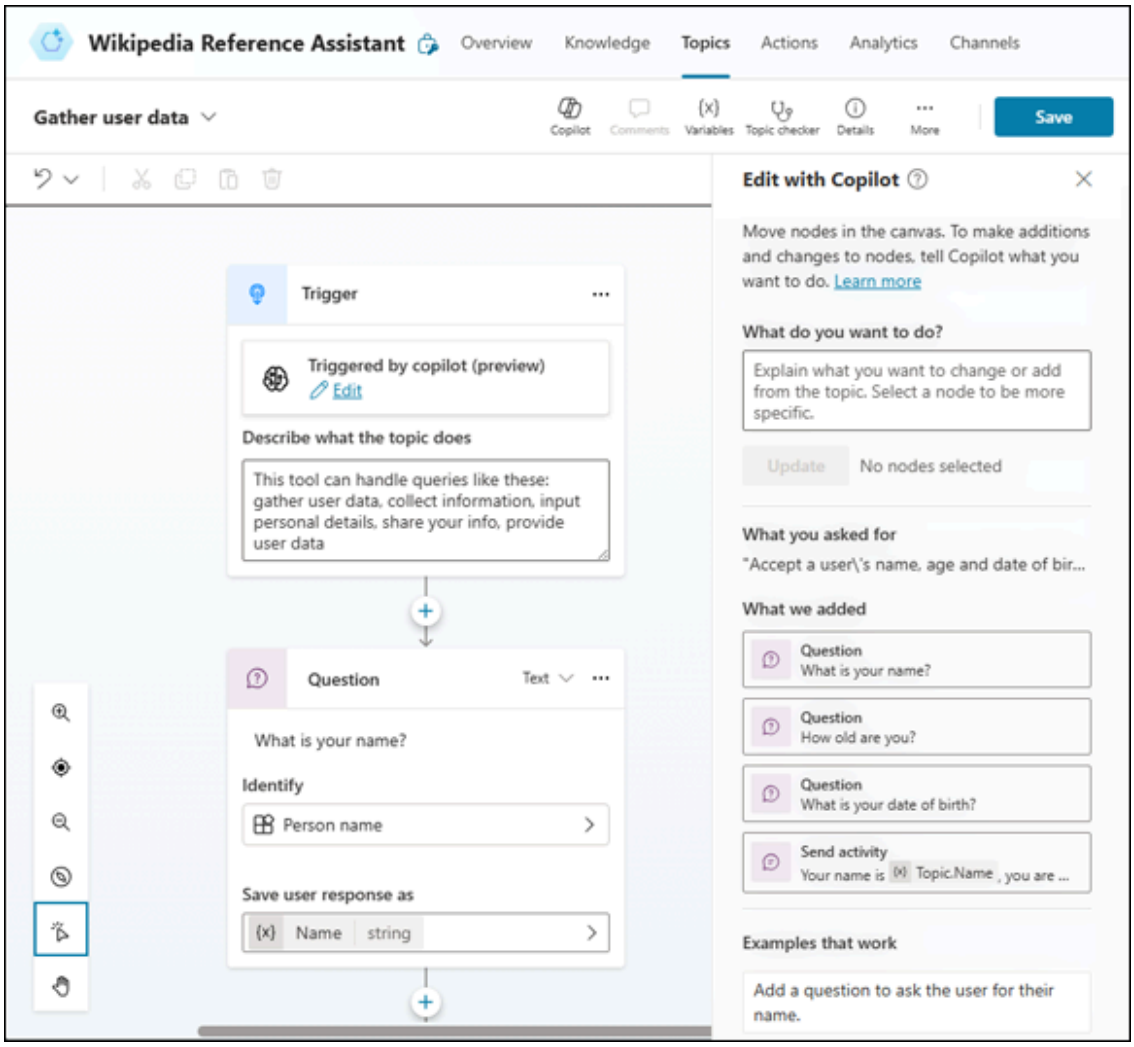


FIGURE 4-19: Editing a new action.

When you're happy with your new action, click the Save button and try it out!

TEST A TOPIC

To test your new topic, first look at the trigger specified by the new topic you just created. In the case of my topic, it uses Copilot to detect whether the user has asked to share their personal details.

If your new topic is working, you should be able to type a prompt into the Test Your Copilot pane in Copilot Studio, and Copilot will activate the topic.

Before you try that, however, you should activate the conversation map. The *conversation map* is a tool that shows you what's happening in your test conversation and why.

To activate the conversation map, click the map icon at the top of the test chat window, as shown in Figure 4-20.

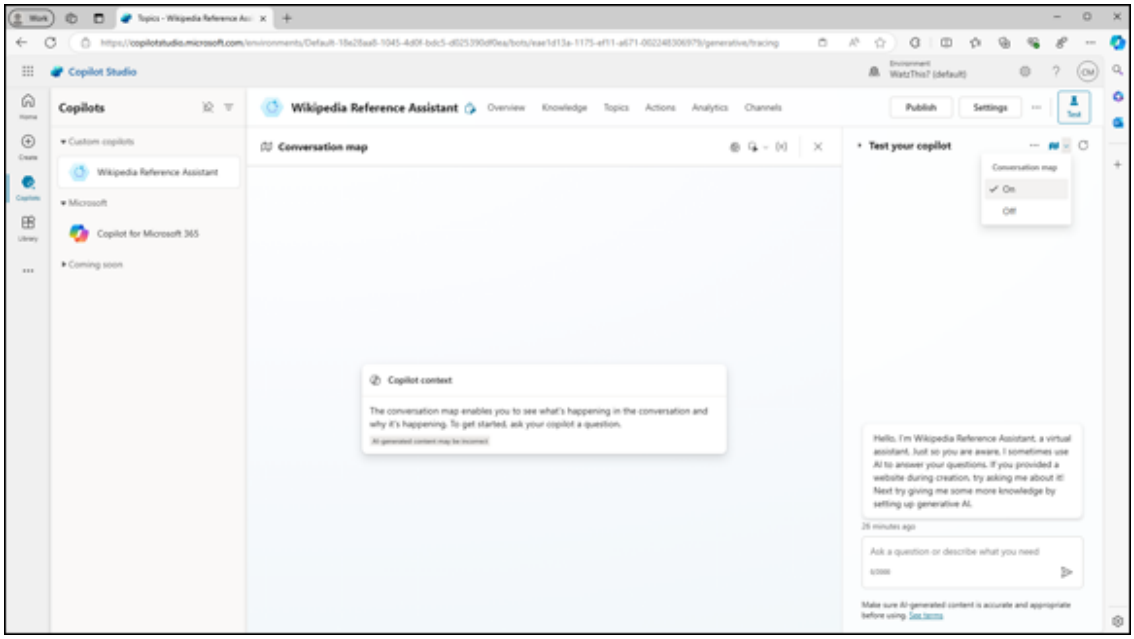


FIGURE 4-20: Opening the conversation map.

When you enable the conversation map, a new window will open in the center area of Copilot Studio, and wait for something to happen in the conversation.

Find the icon at the top of the Conversation map called Track Between Topics and enable it. This icon looks like a speech bubble. Now, it's time to test your agent. Enter a question or other prompt into the test chat window.

Try experimenting with the conversation map's other options and see whether you can trigger other topics. Some prompts you might try include

- Thank you!
- Start a new conversation.
- Hello!
- I want to talk to a person.

Figure 4-21 shows what happens when I prompt my agent with a request to talk to a real person.

The Actions tab

The Actions tab of the Copilot editor is where you can view, create, and edit specific tasks that you want your Copilot agent to do. Actions can be invoked from within topics, or they can be selected and run by Copilot automatically if you have Generative mode turned on for the agent.

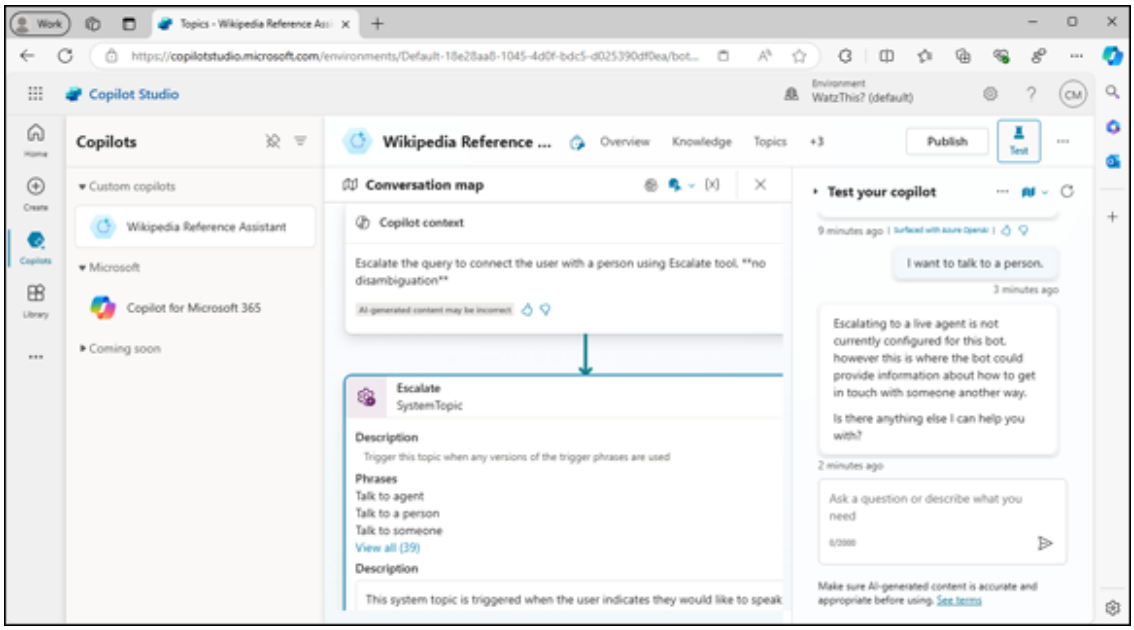


FIGURE 4-21: Asking to talk with a person.



REMEMBER Actions are the things your copilot can do. Topics may consist of triggers, messages, variables, conditions, and actions that are put together into a flow.

Follow these steps to add a weather action to your Copilot agent:

1. **Click the Add an Action button in the Actions tab.**
   The window shown in Figure 4-22 will appear. This window allows you to search for existing actions that you have access to or to create or import a new action.

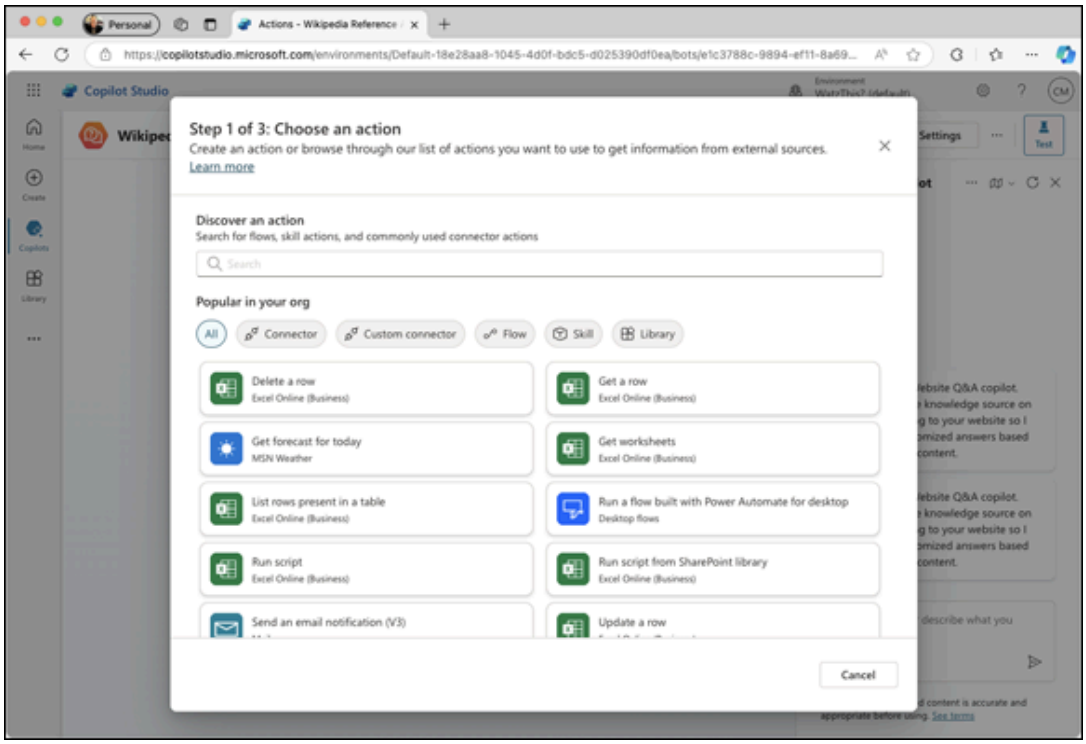


FIGURE 4-22: Choose or create an action.

2. **Find the action named Get Forecast for Today and click on it to start configuring it.**
3. **Choose the User Authentication option in the Connector settings.**
4. **Modify the Action Name, Display Name, or Description if you like. See Figure 4-23.**

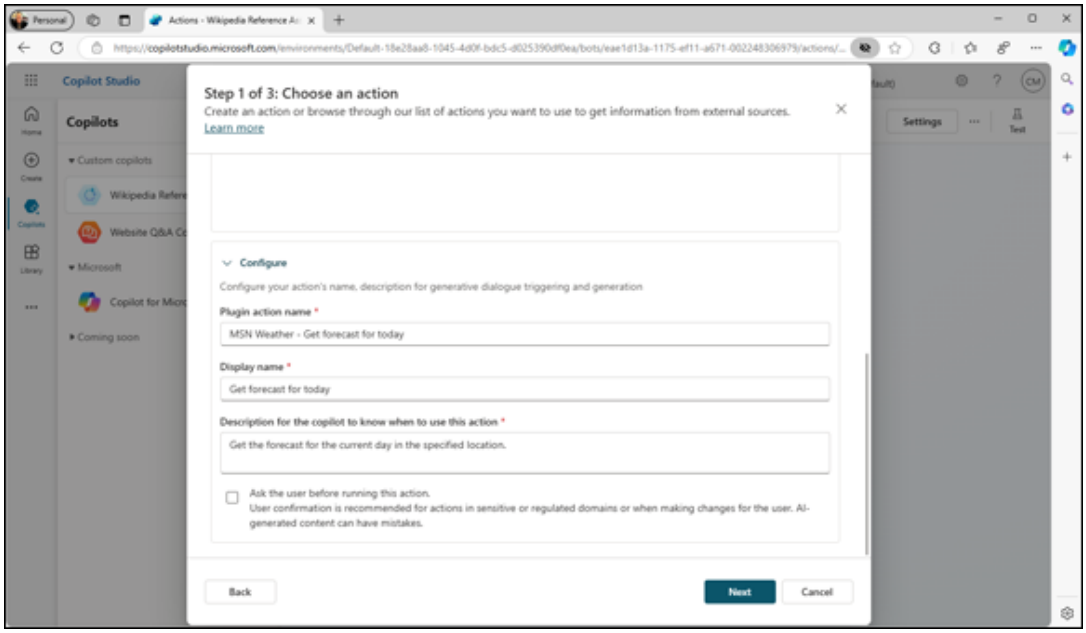


FIGURE 4-23: Configuring the action.

5. **Click Next to view the inputs and outputs for the action, as shown in Figure 4-24.**

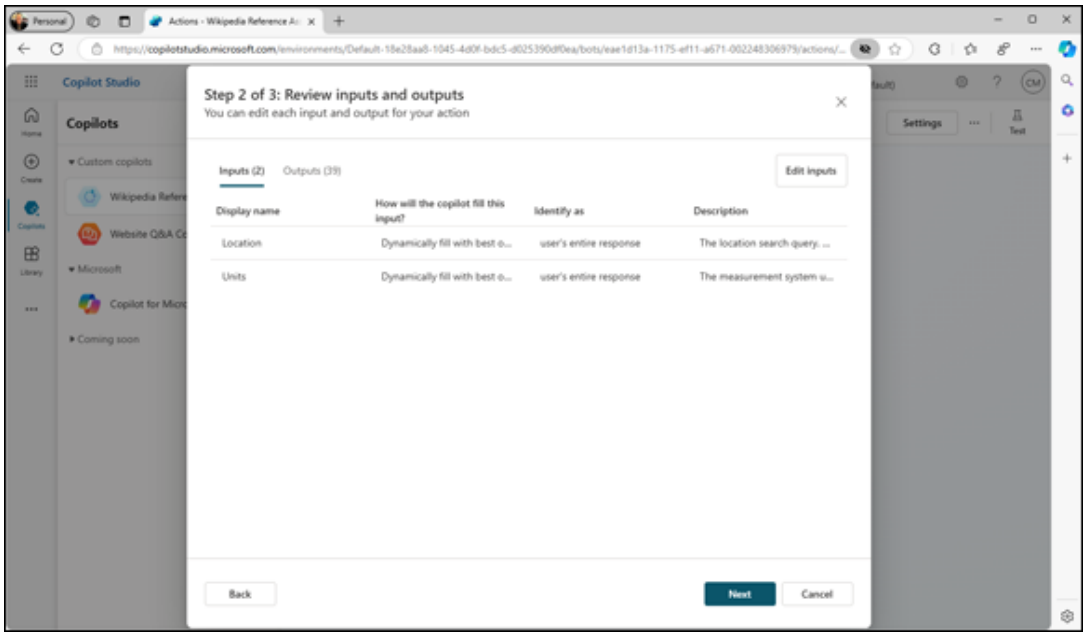


FIGURE 4-24: Viewing inputs and outputs.

6. **Click Next to view the Review and Finish screen.**
   You shouldn't need to change anything here.
7. **Click Finish.**

After a moment, your action will be saved and will appear on the Actions page.

Try out your action by entering something like "What's the weather forecast for Portland, OR?" When I tried this, the action was triggered by Copilot and asked me to specify whether I wanted to use the Imperial or Metric measurement system. After I answered that, my copilot notified me that it needed additional permission to run the action, as shown in Figure 4-25.

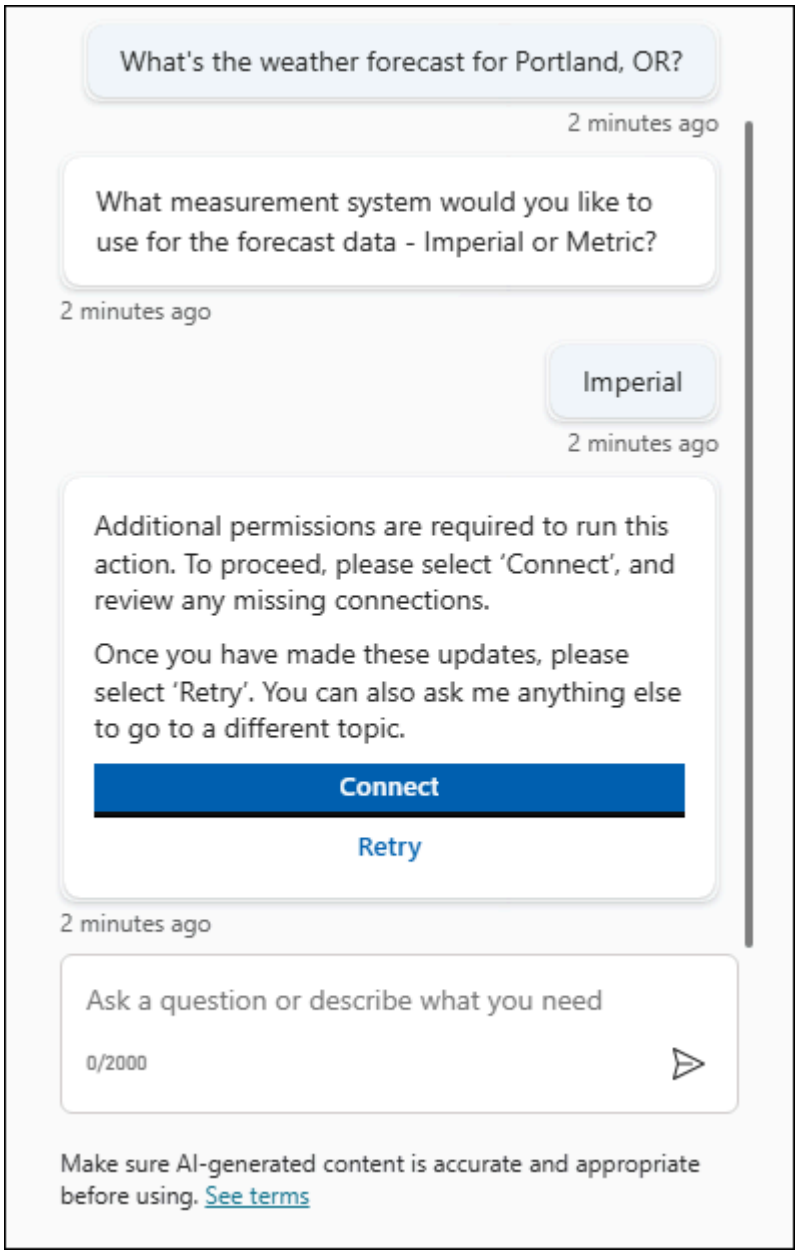


FIGURE 4-25: Copilot needs more permission.

I clicked the Connect button, and a new tab opened in my browser where I could click through a couple of steps to connect the MSN Weather action.

After connecting the action, I went back to the window with my test conversation and retried my prompt.

My copilot didn't respond, but when I switched to the conversation map, I saw that the results did come back from the MSN action, as you can see in Figure 4-26.

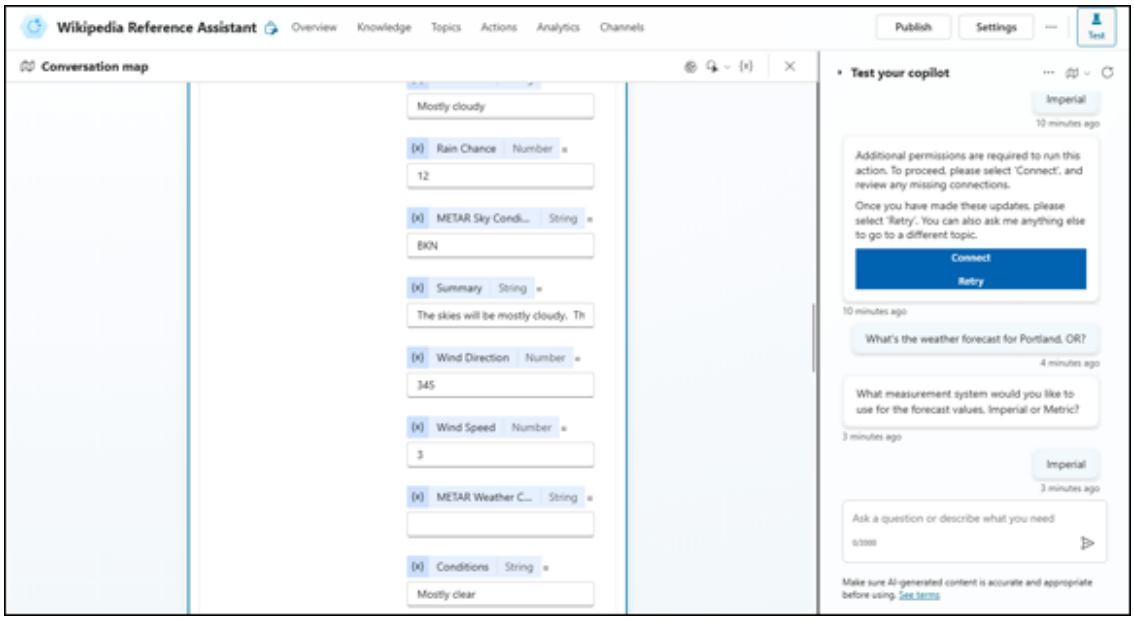


FIGURE 4-26: Viewing the result of an action.

To cause Copilot to report the weather back to you, click the action to open it for editing; then click the Outputs tab and check the box shown in Figure 4-27, which is labeled Respond to the User After Running this Action.

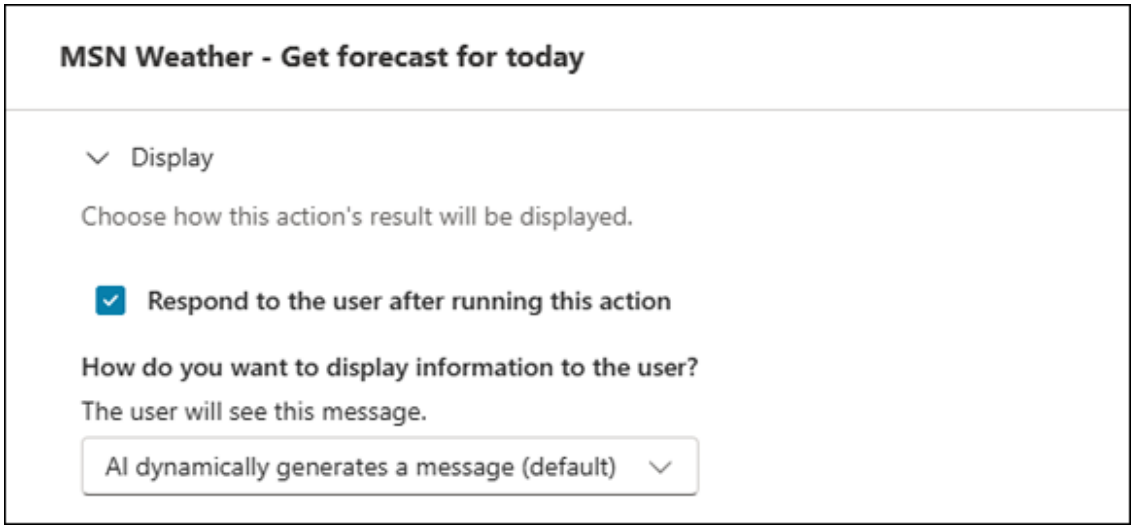


FIGURE 4-27: Enabling your copilot to respond to the user.

Try your prompt again now, and it should respond with an AI-generated forecast using the data from the MSN Weather connector, as shown in Figure 4-28.

The Analytics tab

The next tab in the Copilot Agent Editor is the Analytics tab. When you go to this tab, you see various graphs displaying information about how your agent is being used, as shown in Figure 4-29.

The available data from this screen is particularly important if you've created a customer service agent. Some of the data that's available includes

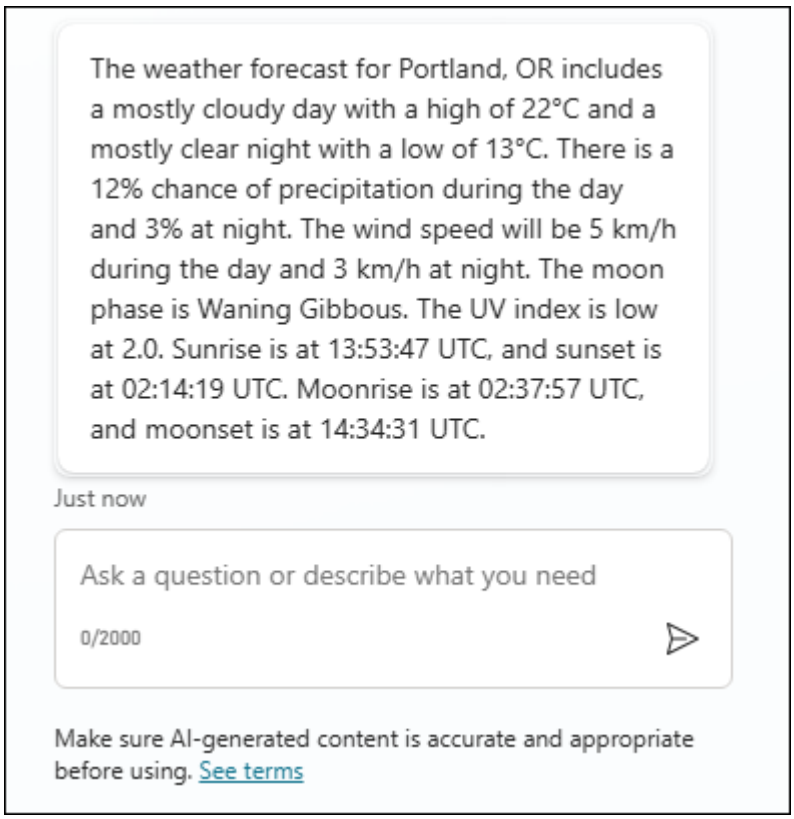


FIGURE 4-28: Getting the weather forecast.

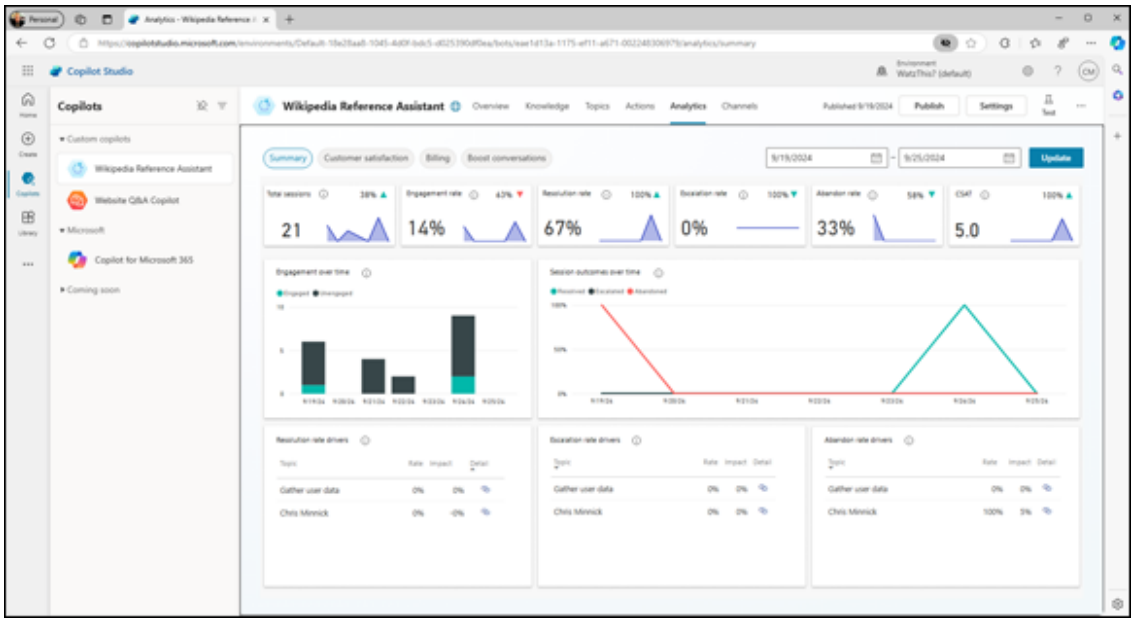


FIGURE 4-29: The agent's analytics interface.

- **Total sessions.** This is the number of conversations that your agent has engaged in during the period of the report.
- **Engagement rate.** This is the percentage of conversations in which the user entered a prompt that triggered a topic to run.
- **Resolution rate.** This is the number of conversations in which the user reaches the "end of conversation" topic. Resolution is typically when the user is asked to respond to a survey.
- **Abandon rate.** This is the percentage of time that an engaged user (someone who enters a prompt that triggers a topic) leaves the conversation without getting to a resolution.

A well-designed and effective agent will have a high resolution rate and a low abandon rate. By monitoring your agent's stats over time, you can find out whether it's doing its job and make adjustments as necessary.

The Channels tab

The last tab is the Channels tab, which is shown in Figure 4-30.

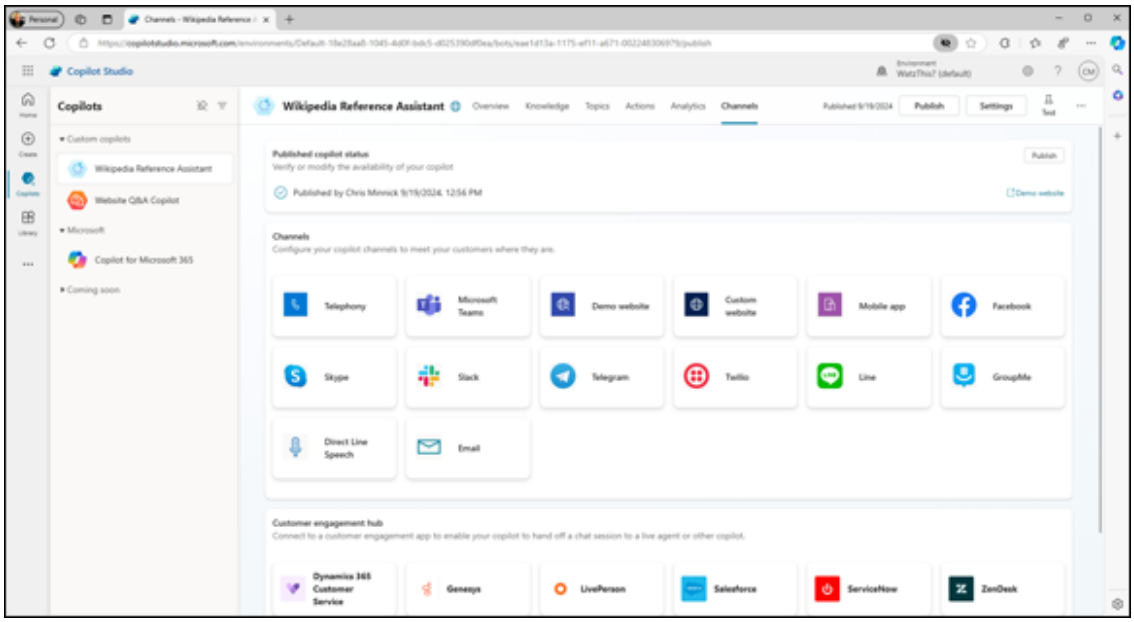


FIGURE 4-30: The Channels tab.

Channels are places where your Copilot agent will be available once it's published. The channels that are currently available are

- **Telephony.** Allows your agent to handle phone calls
- **Microsoft Teams.** Makes your agent available to users in your organization from within Teams
- **Custom Website.** Embeds your agent in a website

- **Custom Mobile App.** Adds your agent to a mobile app
- **Facebook.** Allows users to interact with your agent through Facebook messenger
- **Skype.** Makes your agent available to users via Skype
- **Slack.** Allows people to interact with your agent through Slack
- **Telegram.** Connects your agent to the Telegram messaging app
- **Twilio.** Connects your agent to Twilio, giving it the ability to communicate via text messages
- **Line.** Connects an agent to the Line app, which is a free voice and video calling app
- **GroupMe.** Connects with Microsoft's GroupMe group messaging app
- **Direct Line Speech.** Connects with text-to-speech and speech-to-text capabilities
- **Email.** Allows your agent to interact with users via email

When you click on any of the available channels, a popup window will open that contains instructions and any links necessary to enable that channel. For example, Figure 4-31 shows the code that can be used to embed my Wikipedia assistant agent into a website.

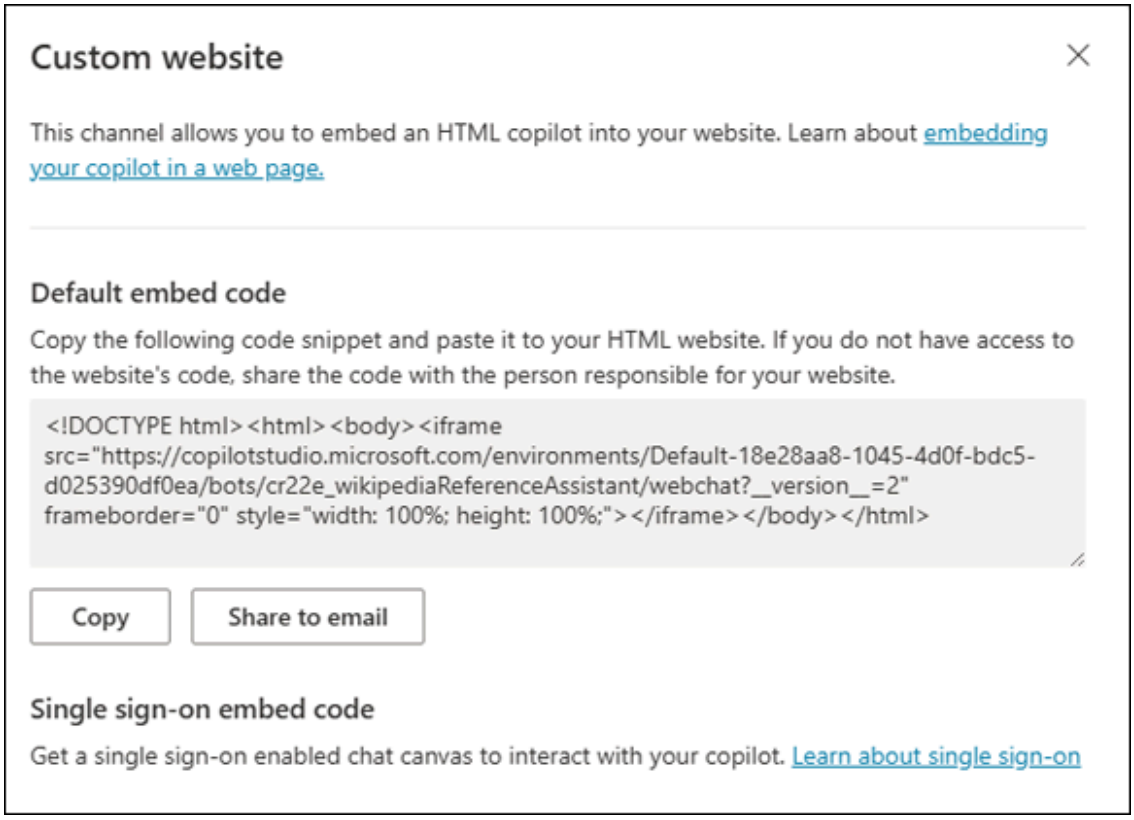


FIGURE 4-31: How to embed the agent in a website.

Before you can use your new agent in your selected channels, you have to publish it. You learn how to publish an agent in the next section.

Publishing Your Agent

When you've finished your agent and are ready to deploy it and test it in the real world, you can publish it. To publish your agent, click the Publish button in the upper-right corner. A popup will display asking you to confirm that you want to make your agent available in the connected channels. When you answer Yes, your agent will be published!

Figure 4-32 shows my Wikipedia Reference Bot embedded in a page on my website at `www.chrisminnick.com/wikibot`.

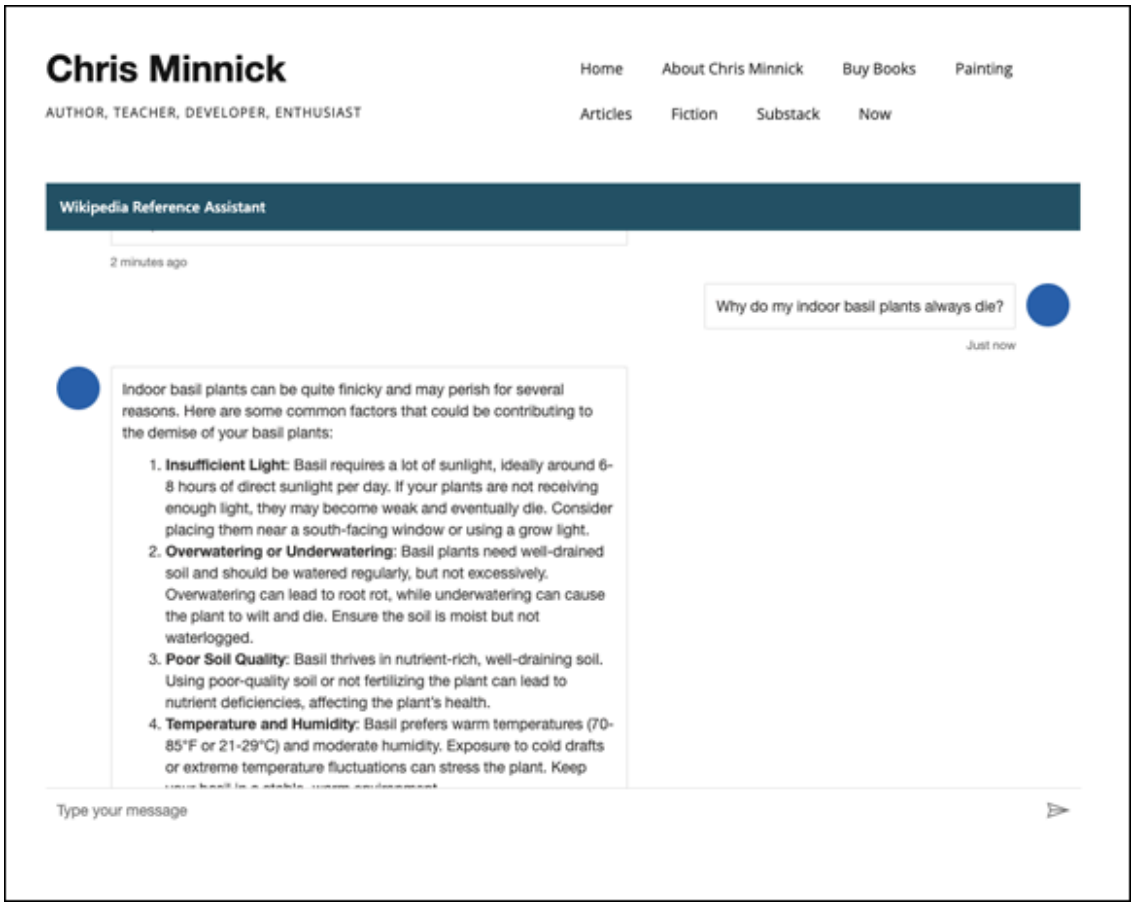


FIGURE 4-32: My Wikipedia Reference Bot.

Exploring Copilot's more advanced features, such as Copilot Studio and custom Copilots, opens up infinite possibilities for how people and companies can use generative AI. With Copilot Studio, you can now have your own AI assistant, with access to your own selected data sources.

In the next chapter, you learn how to use — and even create — Copilot plugins.